



DWF_dp_sat functions

Arithmetic Saturation

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Description

Revision History

The DWF_dp_sat functions truncate the upper bits of argument `a` to the width specified by argument `size` and returns a saturated value if an overflow (or underflow) occurs. Argument `a` and the return value are both either signed (two's complement) or unsigned.

The DWF_dp_sati functions have the same functionality, except that all output bits are inverted.

Table 1-1 Function Names

Function Name	Description
DWF_dp_sat	VHDL unsigned saturate
DWF_dp_sat	VHDL signed (two's complement) saturate
DWF_dp_sat_uns	Verilog unsigned saturate
DWF_dp_sat_tc	Verilog signed (two's complement) saturate
DWF_dp_sati	VHDL unsigned saturate, inverted output
DWF_dp_sati	VHDL signed (two's complement) saturate, inverted output
DWF_dp_sati_uns	Verilog unsigned saturate, inverted output
DWF_dp_sati_tc	Verilog signed (two's complement) saturate, inverted output

Table 1-2 Argument Description

Argument Name	Type	Width / Values	Description
<code>a</code>	Vector	<code>width</code>	Input data
<code>size</code>	Integer	<code>> 1, < width</code>	Word length of return value (VHDL only, constant)
DWF_dp_sat DWF_dp_sati	Vector	<code>size</code>	Return value

Table 1-3 Parameter Description (Verilog)

Parameter	Values	Description
<code>width</code>	≥ 1	Word length of input <code>a</code>
<code>size</code>	<code>> 1, < width</code>	Word length of return value

Verilog Include File: DW_dp_sat_function.inc

Functional Description

```
z[size-1:0] = DWF_dp_sat (a[width-1:0], size)
```

Unsigned Saturate

```
z[size-1:0] = 2size-1      if a[width-1:0] > 2size-1  
              = a[size-1:0]  else
```

Signed Saturate

```
z[size-1:0] = 2size-1-1    if      a[width-1:0] > 2size-1-1  
              = -2size-1    else if a[width-1:0] < -2size-1  
              = a[size-1:0]  else
```

Unsigned Saturate, inverted output

```
z[size-1:0] = ~ DWF_dp_sat (a[width-1:0], size)
```

Signed Saturate, inverted output

```
z[size-1:0] = ~ DWF_dp_sat (a[width-1:0], size)
```

For more information about the DesignWare datapath functions, refer to the topic titled [DesignWare Datapath Functions Overview](#).

Related Topics

- [DesignWare Datapath Functions Overview](#)
- [DesignWare Building Block IP User Guide](#)

VHDL Example

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;
use DWARE.DW_dp_functions.all;
-- DWARE.DW_dp_functions_arith package if IEEE.std_logic_arith is used

entity DWF_dp_sat_test is
  port (a, b, c : in  signed(7 downto 0);
        z       : out signed(7 downto 0));
end DWF_dp_sat_test;

architecture rtl of DWF_dp_sat_test is
begin
  z <= DWF_dp_sat (a * b, 8) + c;
end rtl;
```

Verilog Example

```
module DWF_dp_sat_test (a, b, c, z);

  input  signed [7:0] a, b, c;
  output signed [7:0] z;

  // Passes the parameters to the function
  parameter width = 16;
  parameter size  = 8;

  // add "$SYNOPSYS/dw/sim_ver" to the search path for simulation
  `include "DW_dp_sat_function.inc"

  assign z = DWF_dp_sat_tc (a * b) + c;

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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